

Unit 1 - Lesson 4

Binary Numbers



Computer Science Principles

Warm Up



Retrieve

your knowledge
and ideas and write
it down silently



Pair

up with a neighbor
and talk about your
reflections

Share

your thoughts in a
class discussion



Yesterday you created your own number system using circles and squares.



Discuss:

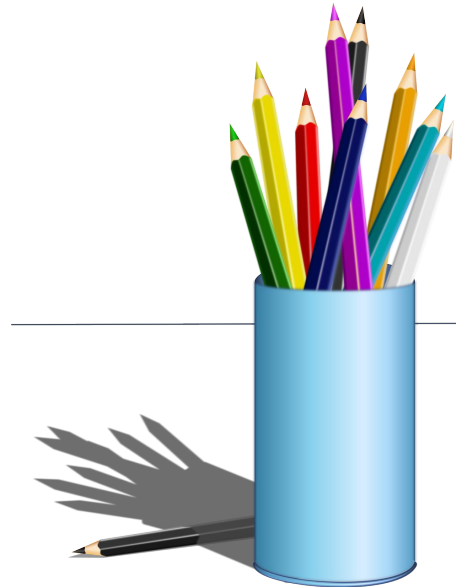
What can we communicate using only two symbols? Is there a limit?

Activity

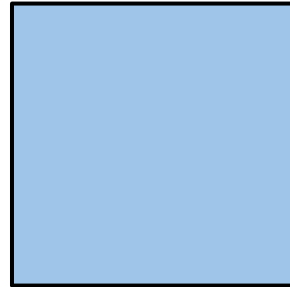
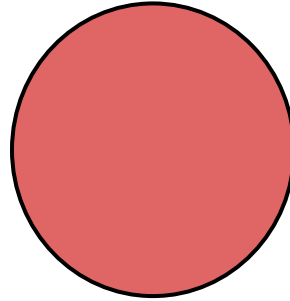


Binary Numbers

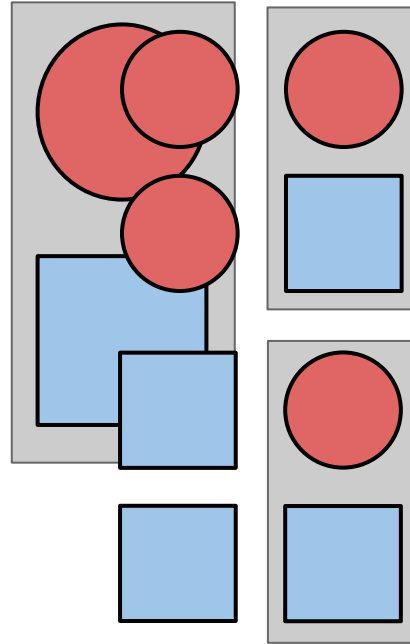
You and your partner should each have:
A Pen or Pencil



1 place value = **2** possible patterns



2 place values= 4 possible patterns



3 place values= 8 possible patterns

We can then map
our patterns to a
numbered list.

0	●	●	●
1	●	●	□
2	●	□	●
3	●	□	□
4	□	●	●
5	□	●	□
6	□	□	●
7	□	□	□

Note: Computer
scientists like to
start counting at 0!





Instead of 2 shapes, what if
we had 10 shapes?

1 place value = Ten 1-shape patterns

0
1
2
3
4
5
6
7
8
9

← These are just shapes!



2 places = one hundred 2-shape patterns

00	10	20	30	40	50	60	70	80	90
01	11	21	31	41	51	61	71	81	91
02	12	22	32	42	52	62	72	82	92
03	13	23	33	43	53	63	73	83	93
04	14	24	34	44	54	64	74	84	94
05	15	25	35	45	55	65	75	85	95
06	16	26	36	46	56	66	76	86	96
07	17	27	37	47	57	67	77	87	97
08	18	28	38	48	58	68	78	88	98
09	19	29	39	49	59	69	79	89	99



Quiz: What comes next?

Ten
shapes

0			
1	<u>0</u>	<u>9</u>	<u>9</u>
2			
3			
4			
5			
6	<u> </u>	<u> </u>	<u> </u>
7			
8			
9			

Quiz: What comes next?

Ten
shapes

0

1 0 9 9

2

3

4

5 1 0 0

6

7

8

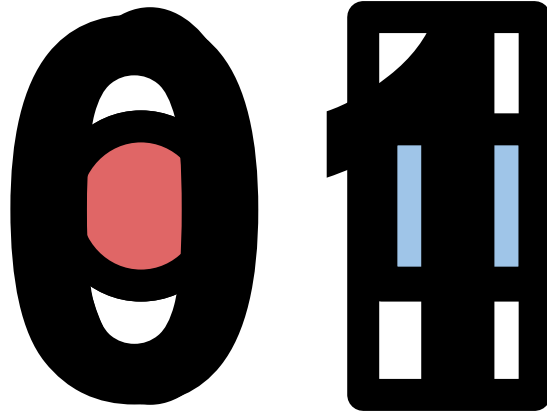
9

Where is this heading?

...binary...



“Binary” is a number system with 2 shapes...



Making Organized Lists -> Counting in Binary

0				0 0 0
1				0 0 1
2				0 1 0
3				0 1 1
4				1 0 0
5				1 0 1
6				1 1 0
7				1 1 1



Make Your Flippy Do!

Flippy Do

Fold along the bold line. Cut on the dotted lines

Name: _____

1. Write in the powers of 2

2. Write in the whole number equivalents

3. Write a row of 0s

4. Write a "1" on the **back** of each flap.
(Careful about upside-down)

5. Cut on dotted lines

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
0	0	0	0	0	0	0	0
				Flip it up!			



Each place value represents one "**bit**" (binary digit). A bit can be a zero or a one.

Your flippy do has 8 bits...

which together make...

1 byte



Try Out Your Flippy Do!

Represent these decimal numbers in binary

- 7 0000 0111
- 20 0001 0100

Represent these binary numbers in decimal

- 0001 0010 18
- 0001 1111 31





Flippy Do Activity Guide

Unit 1 Lesson 4

Name(s) _____ Period _____ Date _____

Activity Guide - Flippy Do Part 1

C

D

E

Directions

Use your Flippy Do or the binary odometer widget to answer the questions.

All 4-Bit Numbers: Fill in the binary equivalents for the decimal numbers below. We've started the first three for you.

Binary: 4-bit number	Decimal
0000	0
0001	1
0010	2

Binary: 4-bit number	Decimal

What do you notice when you compare the odd numbers with the even numbers? What might explain this?

Binary Numbers with exactly one 1: Complete the chart with all 8-bit binary numbers that have exactly one 1. We've done the first two for you.

Binary: 8-bit number (with exactly one 1)	Decimal
0000 0001	1
0000 0010	2
0000 0100	

Binary: 8-bit number (with exactly one 1)	Decimal

What do you notice about the decimal equivalents above?

Computer Science Principles1

Wrap Up

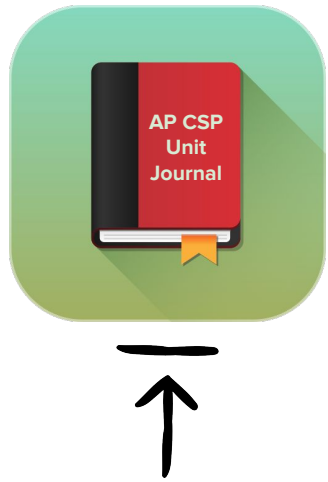




Vocabulary Canvas

Images/Pictures Created
Associations

You will surround the vocabulary word with a design
You ~~also~~ add to your vocabulary as your
that ~~interesting~~ ~~app~~ ~~image~~ ~~that~~ ~~tells~~ ~~the~~ ~~story~~ of
the ~~word~~ ~~itself~~ ~~or~~ ~~the~~ ~~context~~ ~~of~~ ~~the~~ ~~word~~ ~~and~~
something else that helps you to remember (since
they are always a work in progress)
person drawing a mouth with 8 teeth)



Reserve space
in your AP CS
Principles Unit
Journal for a
**Vocabulary
Canvas**

0 to 255

Byte

Binary: 1 0 1 1 0 0 1 1

Made up of bits

eight bits!

64	32	16	8	4	2	1
0	0	0	0	0	0	1
0	0	0	0	0	0	1
0	0	0	1	0	1	0
0	0	0	1	0	1	0

Flip it up!

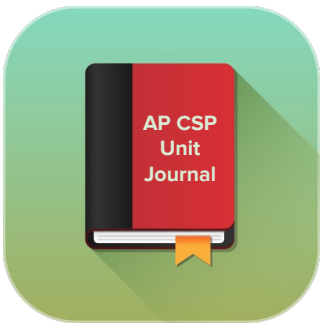
ary Flippy Do

2 2 2 2 2 2 2 2

64 32 16 8 4 2 1

0 0 0 1 0 1 0 1





Vocabulary Canvas

What else can you add?

Links to Outside Resources

<https://en.wikipedia.org/wiki/Byte>

Acronyms

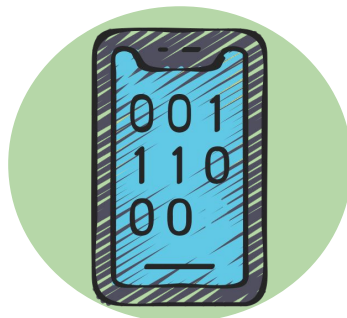
Bits **You** **Total** to **Eight**

The Socials



Byte

Quick Sketches



Bad Jokes / Puns

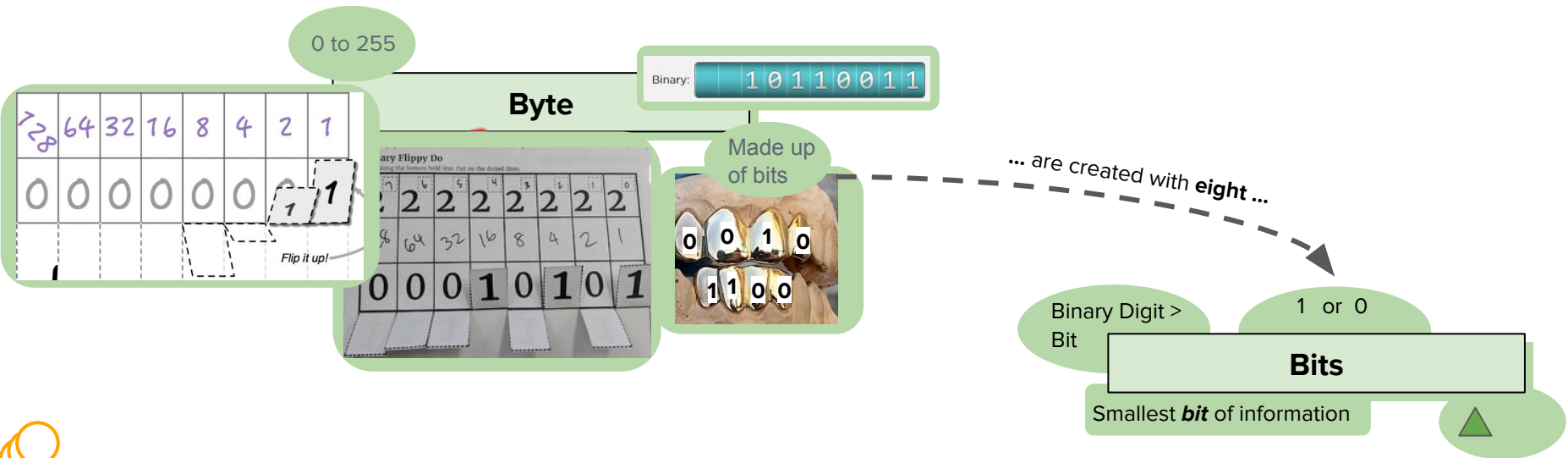
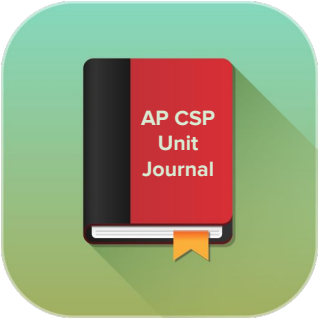
My dog took a **byte**
out of my hard drive.
Now it's in **bits**!

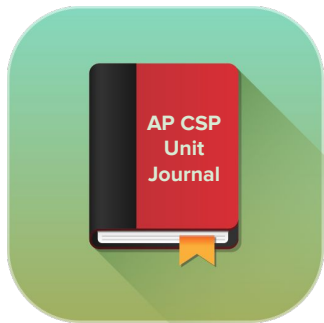


Vocabulary Canvas

Link to Other Words

Create a **concept map** by linking your Canvas to explain the larger connection between vocabulary words and how they interact.





Vocabulary Canvas

Be Creative! Share with Classmates!

Ultimately, a Vocabulary Canvas is designed by you in order to speak to you.

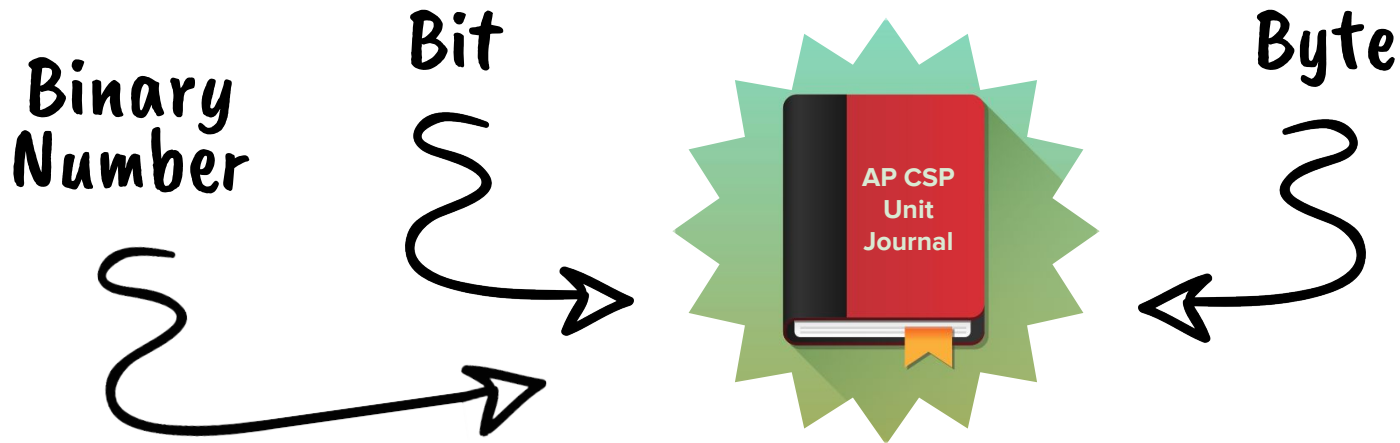
As your understanding grows, so should the Canvas. It's never too late to go back and add an interesting thought or an insightful idea.

*We will use the **Give One, Get One** activity in class in order to share your creations with your classmates.*

Create your Canvas!

Let's add the following *four* words into our **AP CS Principles Unit Journal**. Create a vocabulary page within your journal, or use our digital template.

Start with the **words**, your **initial definitions**, and **visual representations** of the words.



Decimal number: a base 10 number with ten possible different digits

0 1 2 3 4 5 6 7 8 9

10^1	10^0
10	1
2	3

Same number represented two different ways.



Decimal

Binary



Binary number: a base 2 number with two possible different digits

0 1

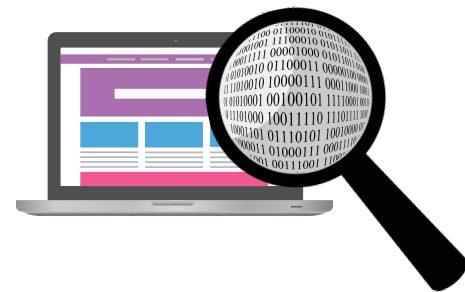
2^4	2^3	2^2	2^1	2^0
16	8	4	2	1
1	0	1	1	1



Prompt: Now that we've had a chance to practice, let's find out what we've learned and what we still have questions about. Write down:

- 3 things you learned today
- 2 things you found interesting
 - 1 question you still have

Bit: A contraction of "Binary Digit"; the single unit of information in a computer, typically represented as a 0 or 1



Byte: 8 bits

10010101